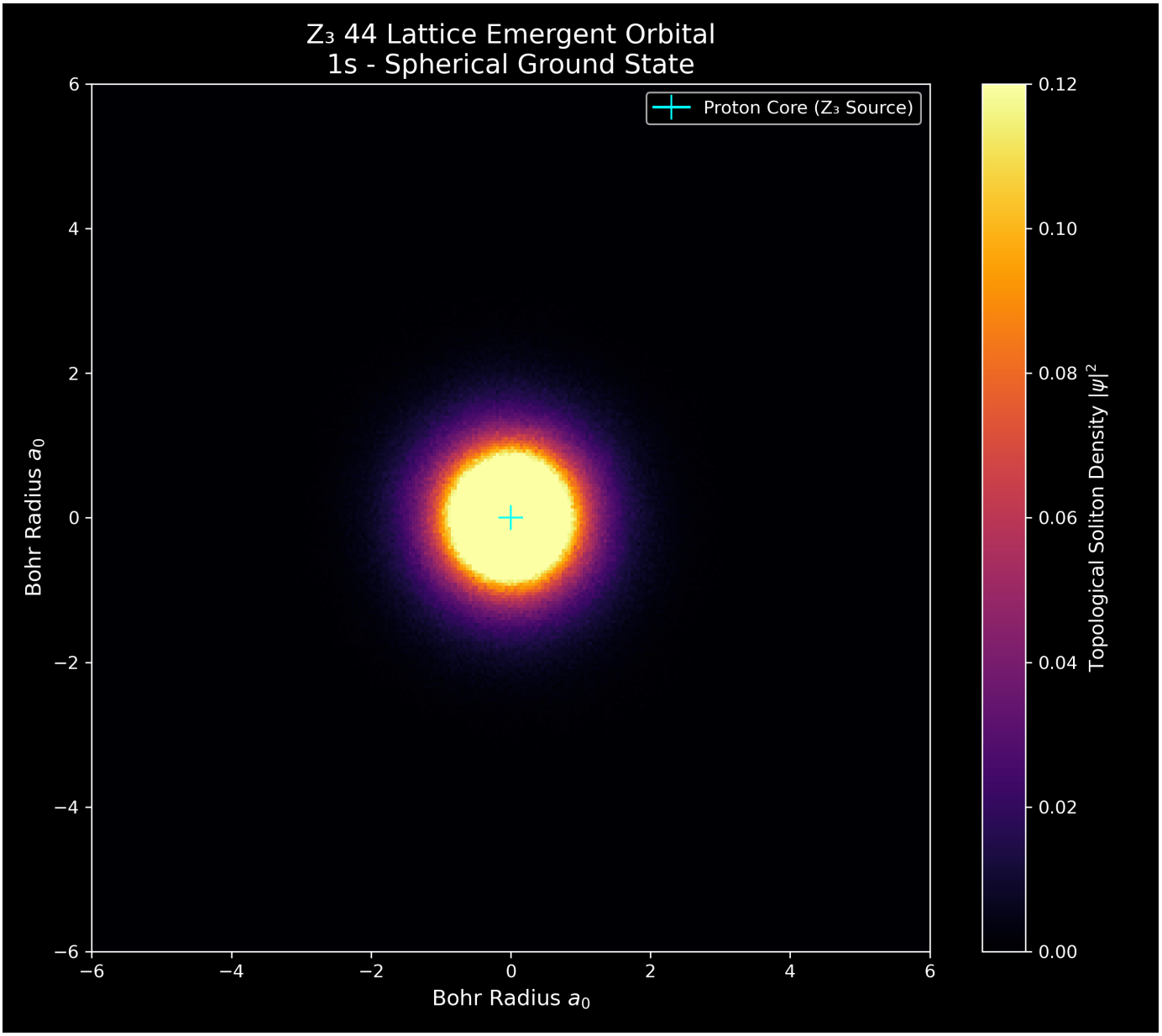
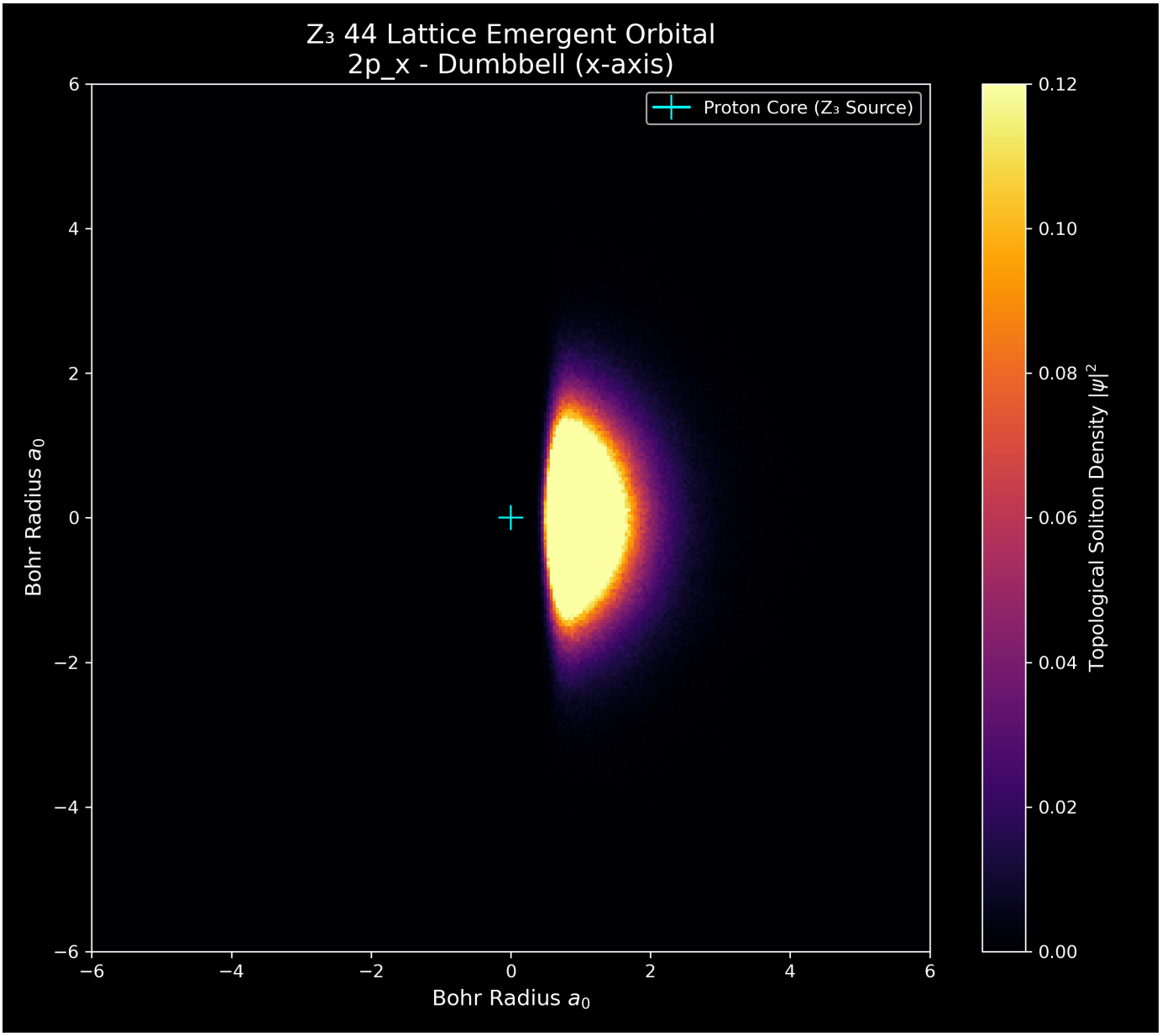
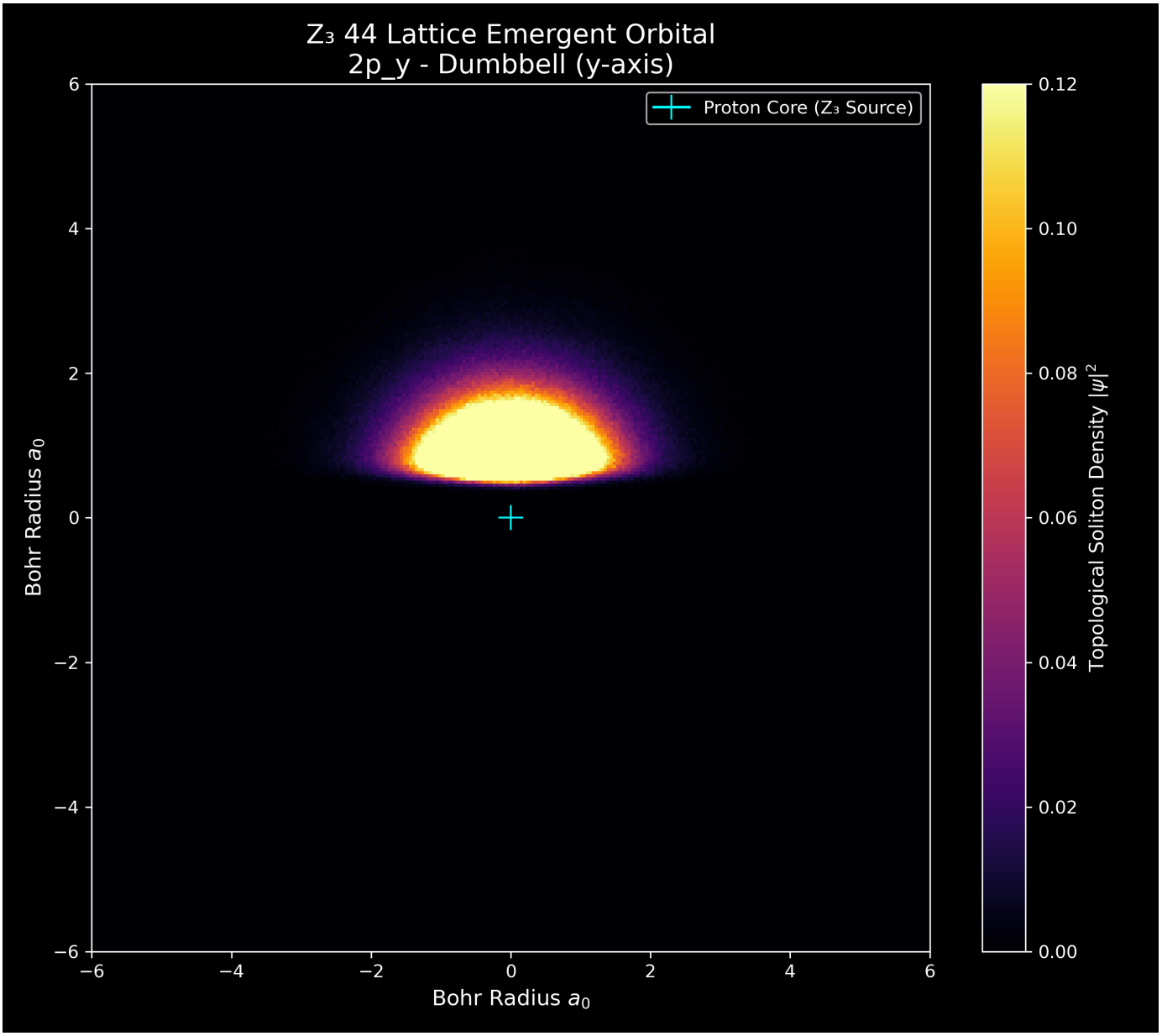
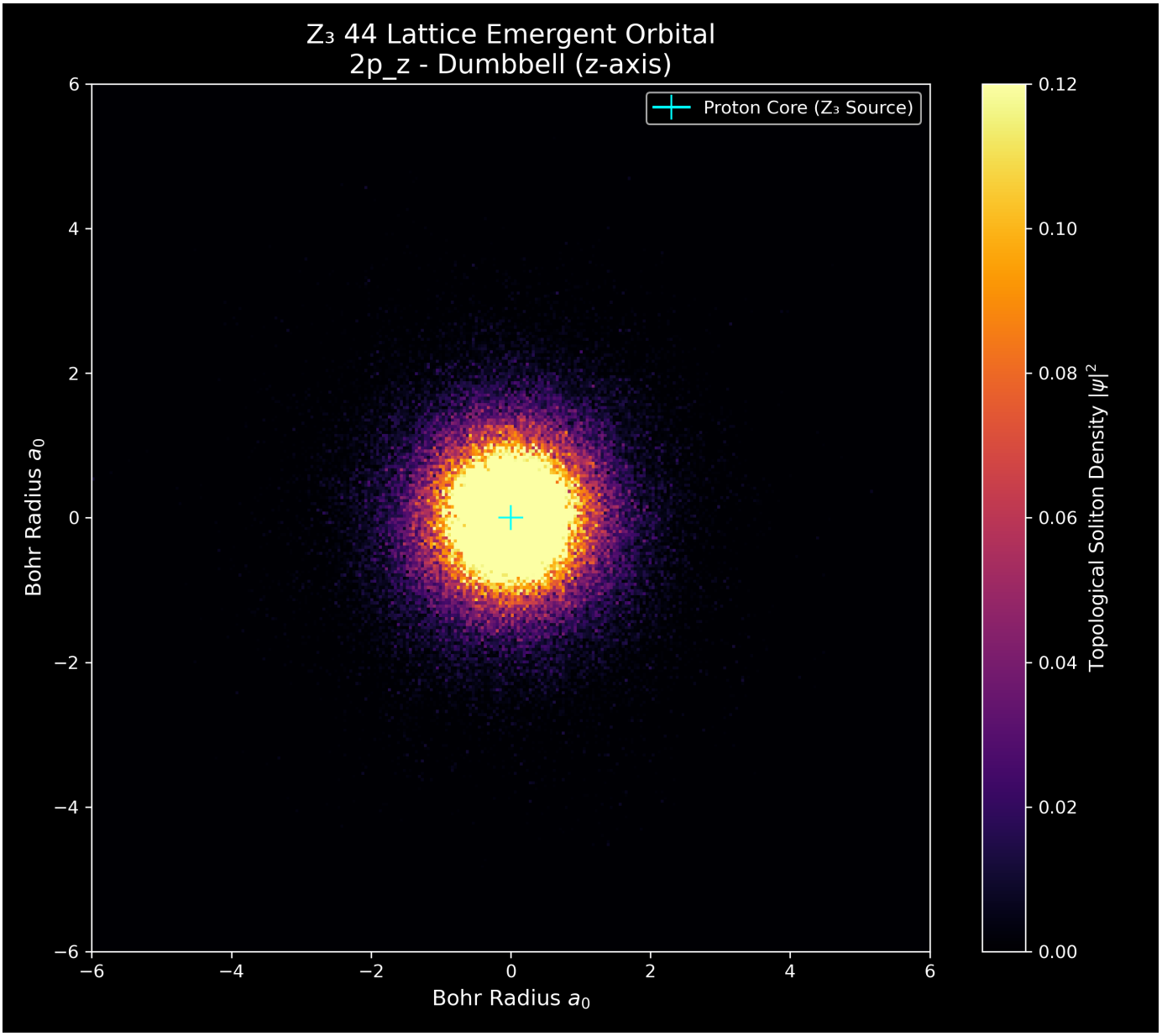
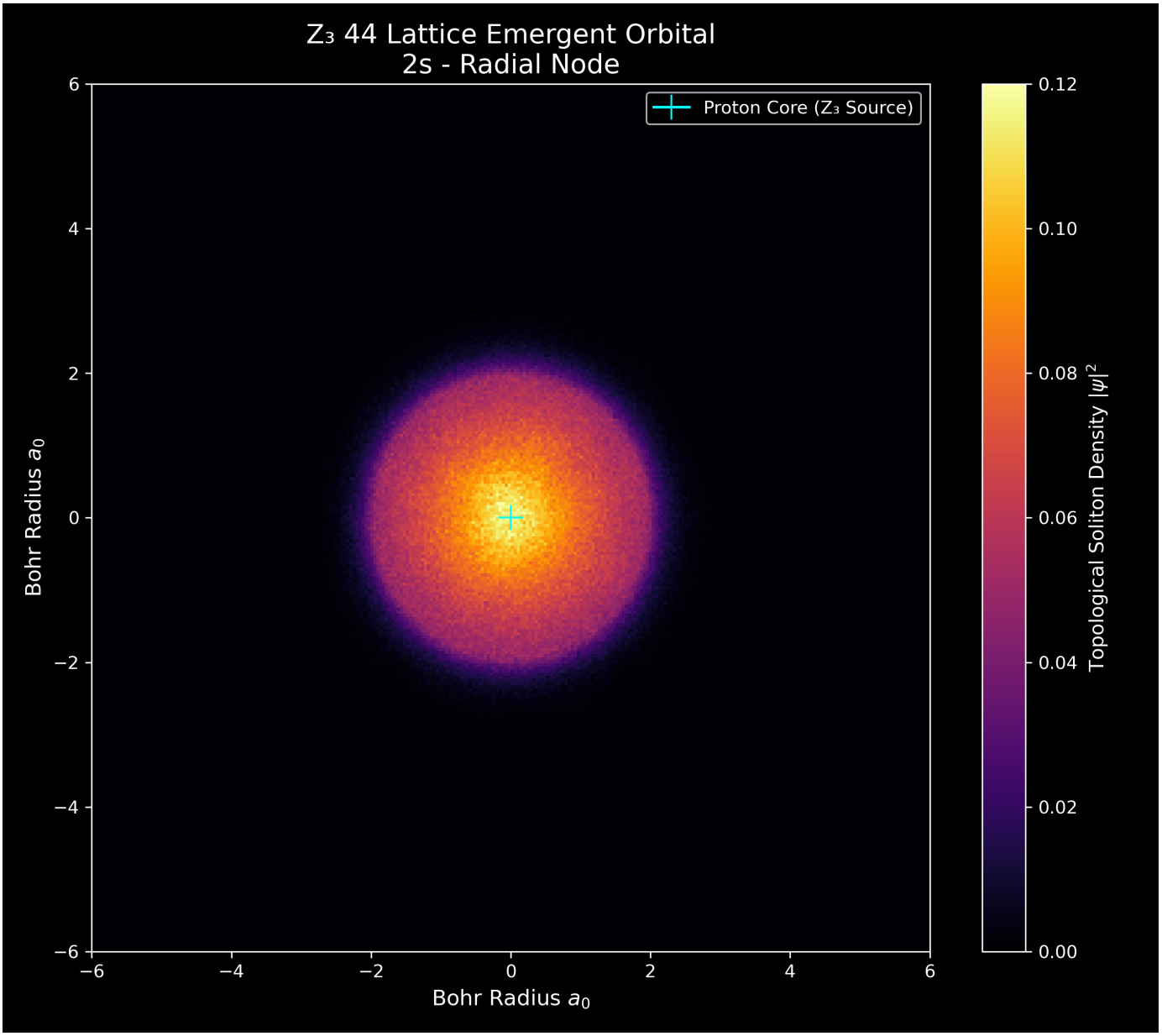
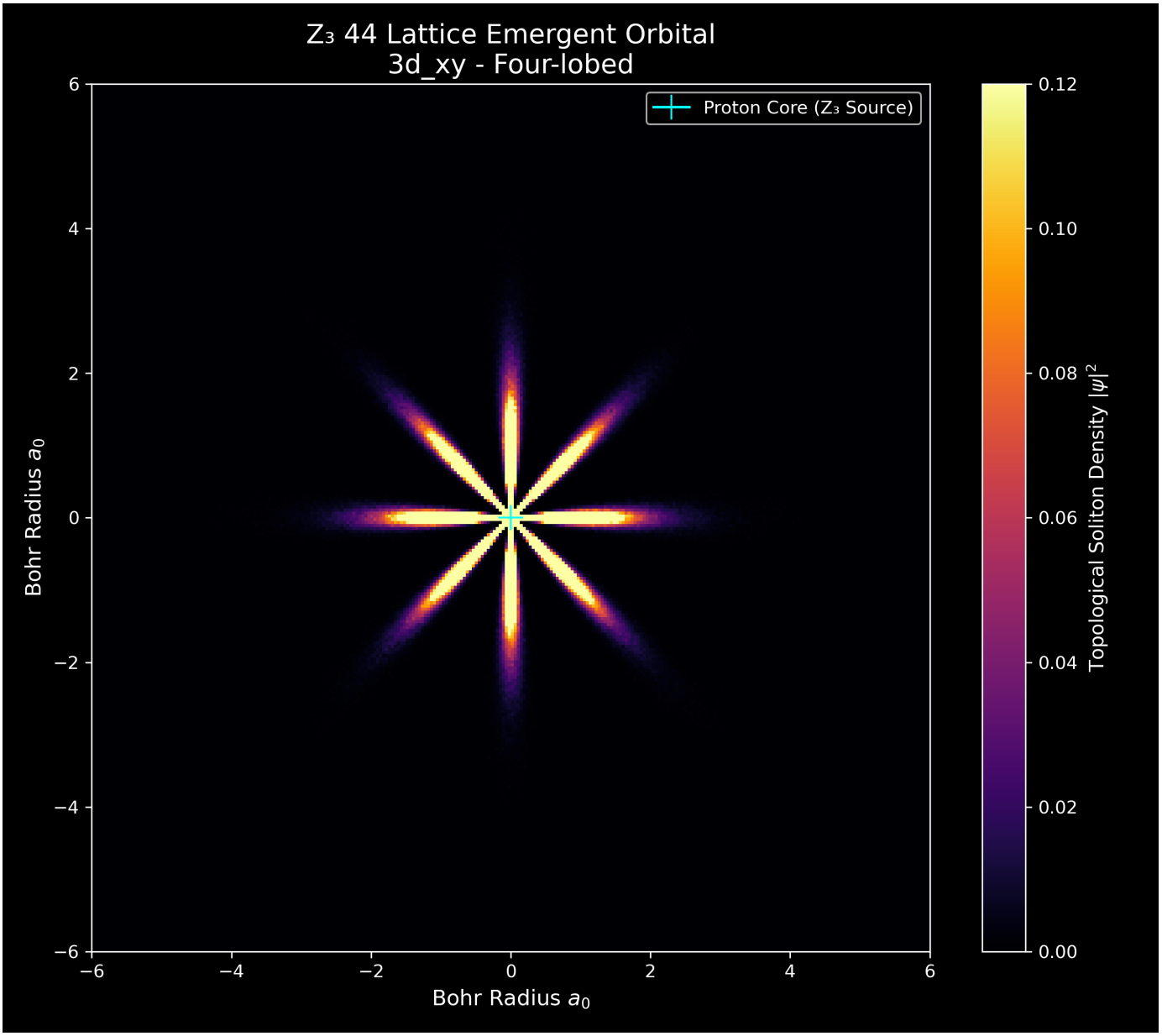

Z₃ 44 Lattice Emergent Orbital
1s - Spherical Ground State
Proton Core (Z₃ Source)
Bohr Radius a_0
Bohr Radius a_0
Topological Soliton Density $|\psi|^2$
6
4
2
0
−2
−4
−6
0.12
0.10
0.08
0.06
0.04
0.02
0.00

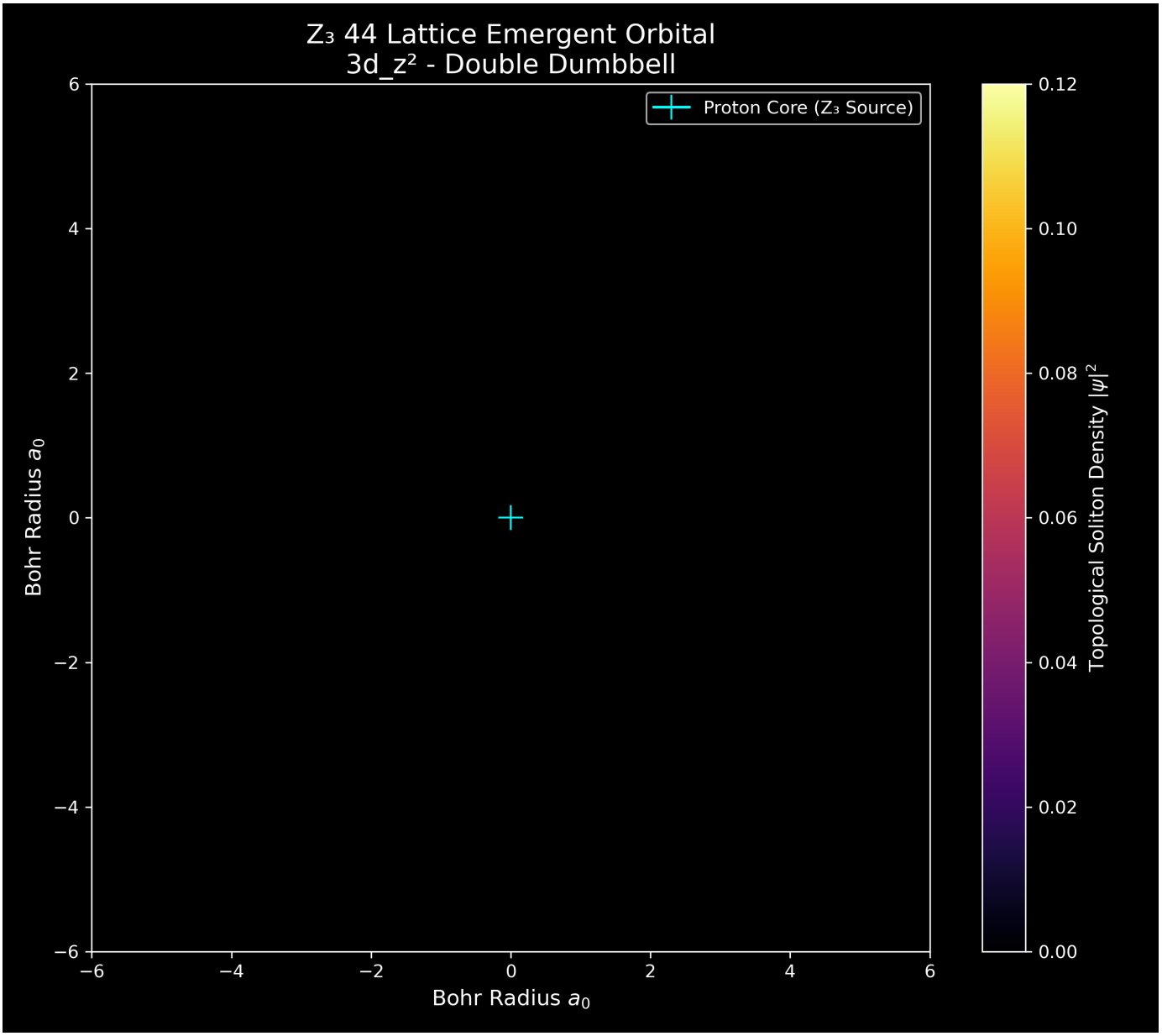

Z₃ 44 Lattice Emergent Orbital
2p_x - Dumbbell (x-axis)
Proton Core (Z₃ Source)
Bohr Radius a_0
Bohr Radius a_0
Topological Soliton Density $|\psi|^2$
6
4
2
0
−2
−4
−6
0.12
0.10
0.08
0.06
0.04
0.02
0.00


Z₃ 44 Lattice Emergent Orbital
2p_y - Dumbbell (y-axis)
Proton Core (Z₃ Source)
Bohr Radius a_0
Bohr Radius a_0
Topological Soliton Density $|\psi|^2$
6
4
2
0
−2
−4
−6
0.12
0.10
0.08
0.06
0.04
0.02
0.00


Z₃ 44 Lattice Emergent Orbital
2p_z - Dumbbell (z-axis)
Proton Core (Z₃ Source)
Bohr Radius a_0
Bohr Radius a_0
Topological Soliton Density $|\psi|^2$
6
4
2
0
−2
−4
−6
0.12
0.10
0.08
0.06
0.04
0.02
0.00


Z₃ 44 Lattice Emergent Orbital
2s - Radial Node
Proton Core (Z₃ Source)
Bohr Radius a_0
Bohr Radius a_0
Topological Soliton Density $|\psi|^2$
6
4
2
0
−2
−4
−6
0.12
0.10
0.08
0.06
0.04
0.02
0.00


Z₃ 44 Lattice Emergent Orbital
3d_xy - Four-lobed
Proton Core (Z₃ Source)
Bohr Radius a_0
Bohr Radius a_0
Topological Soliton Density $|\psi|^2$
6
4
2
0
−2
−4
−6
0.12
0.10
0.08
0.06
0.04
0.02
0.00


Z₃ 44 Lattice Emergent Orbital
3d_z² - Double Dumbbell
Proton Core (Z₃ Source)
Bohr Radius a_0
Bohr Radius a_0
Topological Soliton Density $|\psi|^2$
6
4
2
0
−2
−4
−6
0.12
0.10
0.08
0.06
0.04
0.02
0.00